

# Gradient Boosting for Pollution Mortality Prediction

60 observations | 15 predictors | 10-fold CV | final test RMSE 34.68

## Dataset

60 observations and 15 input variables  
Target: age-adjusted mortality rate MORT  
Climate, pollution, and social context

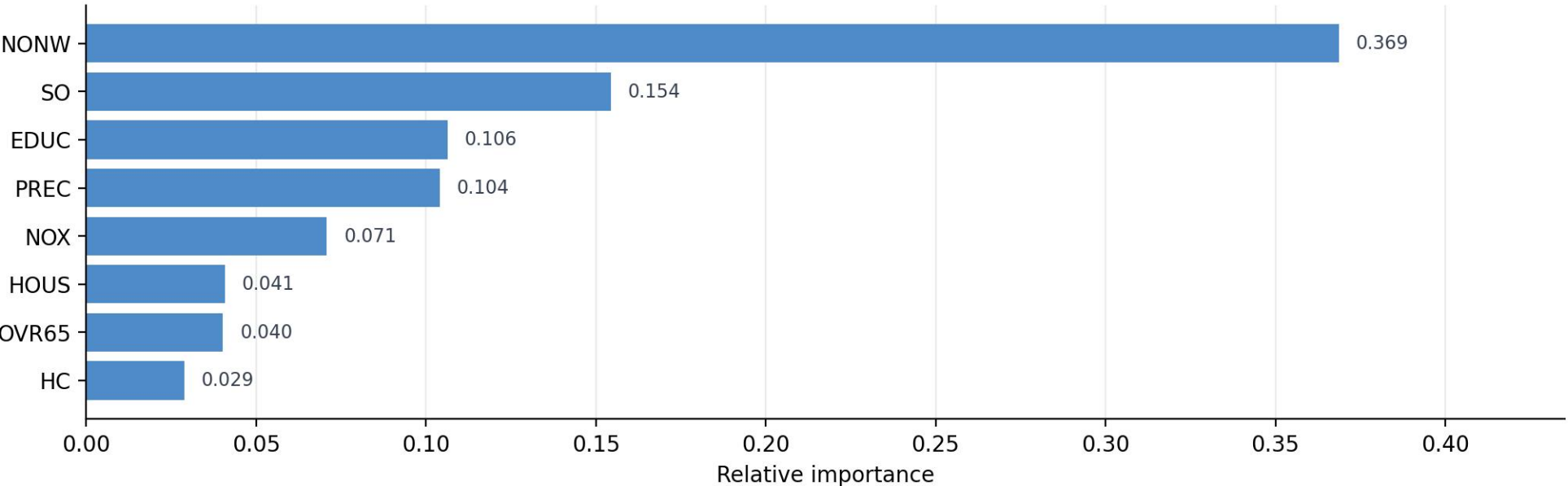
## Workflow

Inspect distributions and skewed variables  
Split train, validation, and test data  
Train Gradient Boosting and tune with CV  
Rank variables by predictive importance

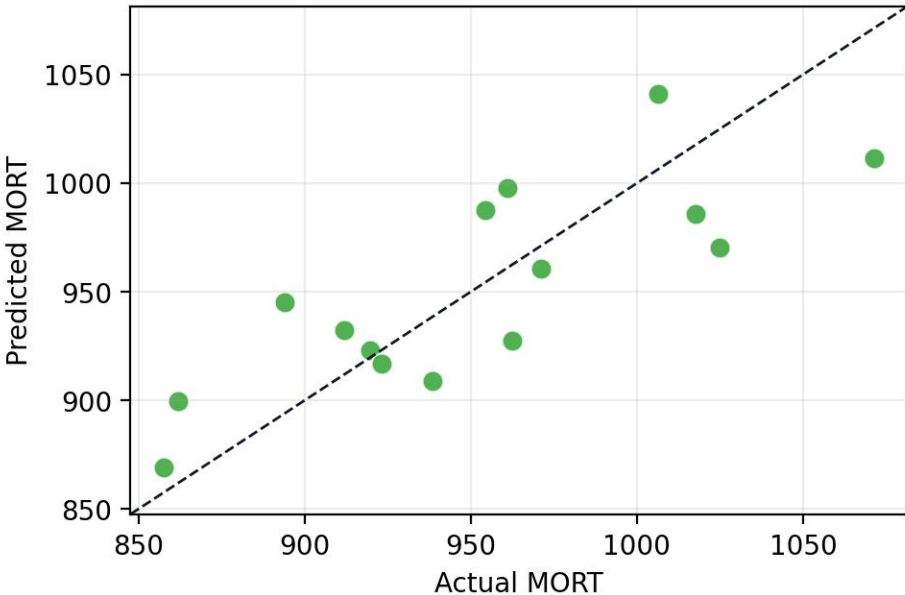
## Overfitting Signal

500 depth-3 trees memorize the training split  
Train RMSE = 0.00  
Validation RMSE = 54.34  
Test RMSE = 41.82  
Training error alone is misleading

## Top Feature Importances



## Test Predictions



## Best CV Model

$n_{\text{estimators}} = 50$   
learning\_rate = 0.1  
max\_depth = 2  
CV RMSE = 38.84  
Train RMSE = 10.08  
Test RMSE = 34.68

## Interpretation

Top variables: NONW, SO, EDUC, PREC, and NOX  
Social variables may proxy inequality and urban structure  
Pollution and climate capture exposure context  
Importance is predictive, not causal

## Recommendations

Use cross-validation for model choice  
Report train, validation, and held-out error  
Try log-pollution and interaction features  
Validate on more cities before policy claims